

TEXTILE MODULE — STEP-BY-STEP GUIDE

A worked, testable example (company: DEMO)

This guide walks the whole textile flow — **knitting → dyeing → finishing → sell** — one step at a time, with real example values. Every figure below is produced by the built-in demo, so you can open each screen in the **DEMO** company and see the exact same numbers. The same command also loads **~10 hand-test records in every module**, spread across their stages (see Part H): `php artisan db:seed --class="Database\Seeders\DemoWalkthroughSeeder"`

THE BIG PICTURE

STAGE	WHAT HAPPENS
Master data (once)	Set up process types, machines, the fabric specification (with its material recipe) and lab dip (colour recipe).
Plan	Turn a customer order into a Production Plan (Knit → Dye → Finish) with dates.
Knit	In-house or sub-contract. Yarn → grey fabric. Materials auto-calculated from the recipe.
Dye	Grey + dyes + chemicals → dyed fabric, against an approved lab dip. Chemicals auto-calculated.
Finish	Dyed → finished fabric. Sold to the customer.

Everything is under the **Textile** menu. Stock and accounts update automatically at each step.

PART A — MASTER DATA (SET UP ONCE)

A1. Fabric specification & recipe. Textile → Fabric Specifications → New.

STEP	WHERE	EXAMPLE VALUES
1	Pick the product	Grey Fabric (Single Jersey)
2	Fabric fields	80% Cotton 20% Polyester · 280/290 GSM · 72" Open · Grey
3	Knitting parameters	Machine dia 30 · Gauge 24 · Stitch length 4.1+5.0+0.5 · Yarn 30s · Combed
4	Material consumption	Cotton Yarn 30s — 1.0 per unit , wastage 5% (knitting loss)

Result: picking this fabric on a knitting order auto-fills all these fields, and its material need is calculated from the recipe. Print it with *Spec sheet* to hand to the machine operator.

A2. Lab dip (colour recipe). Textile → Lab Dips → New, then approve it (Submit → Send to lab → Internal approve → Customer approved).

STEP	WHERE	EXAMPLE VALUES
1	Colour	Navy Blue · Pantone 19-3832
2	Dye-house parameters	Reactive · liquor ratio 1:8 · 60°C · shade 3%
3	Dyeing recipe	Reactive Dye 3% owf · Salt 40 g/L · Soda Ash 15 g/L
4	Approve	Only an approved lab dip can be used on a dyeing order. Print with <i>Recipe sheet</i> .

A3. Dyeing specification (standard dyeing program). Textile → Dyeing Specifications → New. The per-fabric dyeing standard (the parallel of the fabric spec, for the dye house): pick the dyed-fabric product, set the process parameters (process, liquor ratio, temperature, shade, fastness) and the recipe (dyes % owf, chemicals g/L). It auto-fills a dyeing order and prints as a *Recipe sheet*. Roles: the **dyeing specification** is the standard program per fabric; the **lab dip** is the shade approval and can override it for a specific colour.

PART B — MASTER PRODUCTION SCHEDULE (TIME & ACTION)

The plan is the master schedule for an order. It carries the full specification a planner works from, in four blocks:

BLOCK	FIELDS
Order & buyer	Buyer, Style/Article no, Buyer PO no, Order qty + unit (e.g. 1,200 PCS), Colour
Fabric specification	Composition, GSM, Width/Dia, Fabric type, Production qty (KG)
Schedule (T&A)	Booking, Plan, Start, Delivery/Due, Shipment dates
Stages	Knitting → Dyeing → Finishing, each with machine + planned start/end + qty

STEP	WHERE	EXAMPLE & WHAT HAPPENS
1	Sales → Sales Orders → (confirmed order) → <i>Production plan</i>	Creates PLAN-0002 ; buyer, colour and fabric spec auto-fill from the product's specification; stages Knitting → Dyeing → Finishing dated to the delivery date.
2	Textile → Production Plans → open it	Fill buyer/style/PO, adjust dates & machines per stage. Progress shows 0% until runs are recorded, then rolls up automatically.
3	On a stage → <i>Generate process order</i>	Creates the actual run for that stage, linked back to the plan and the order.
4	<i>Plan sheet</i>	Prints the whole schedule (all specs + dated stages) to hand to the floor.

PART C — IN-HOUSE KNITTING

STEP	WHERE	EXAMPLE & WHAT HAPPENS
1	Textile → Process Orders → New	Process = Knitting; output = Grey Fabric (spec auto-fills); planned 480 kg .
2	Inputs → <i>Calculate from recipe</i>	Fills Cotton Yarn = 480 × 1.0 × 1.05 = 504 kg (incl. 5% loss).
3	<i>Issue materials</i>	Yarn leaves stock into Work-in-Progress (WIP).
4	<i>Add costs</i>	Labour 6,000 · machine 12 h · utility 1,500 · overhead 1,000 — capitalised into WIP.
5	<i>Record production</i>	Produced 480, wastage 20 — a tracked grey-fabric batch is received at true cost.
6	<i>Record QC</i>	Passed 470 / rejected 10 — rejects removed from stock; order Completed.

PART D — KNITTING SUB-CONTRACT (JOB-WORK)

STEP	WHERE	EXAMPLE & WHAT HAPPENS
1	Textile → Knitting Sub-contracts → New	Sub-contractor = Acme Knitting Mills ; output = Grey Fabric; service item = Service Charge for Knitting.
2	Service charge	Bill qty 2,000 KG × rate 25 = Tk 50,000 (BDT).
3	<i>Issue yarn</i>	2,000 kg yarn issued to the knitter (into WIP).
4	<i>Record knitting charge</i>	Posts Tk 50,000: Dr WIP / Cr Payable — Acme Knitting Mills . The mill now owes the knitter.
5	<i>Receive grey fabric</i>	1,960 kg grey received at yarn + knitting cost (≈ Tk 280.61/kg), traced to the yarn lot.

STEP	WHERE	EXAMPLE & WHAT HAPPENS
6	Accounts → pay the supplier	Settle Acme Knitting Mills through the normal supplier-payment flow.

Print the *Work order* to hand to the sub-contractor (yarn supplied + spec + service charge).

PART E — DYEING (WITH THE LAB-DIP GATE)

STEP	WHERE	EXAMPLE & WHAT HAPPENS
1	Textile → Process Orders → New	Process = Dyeing; output = Dyed Fabric; approved lab dip = Navy Blue (colour + recipe auto-fill); planned 460 kg.
2	Inputs → add grey fabric, then <i>Calculate from recipe</i>	Adds the chemicals for 460 kg at liquor 1:8 (see Part G).
3	Issue → Add costs → Produce → QC	Same flow as knitting; dyeing cannot start without the approved lab dip.

PART F — FINISHING & SALE

Run a **Compacting/Finishing** process order (dyed → finished fabric), then the finished fabric feeds the customer sales order. Open any finished batch and click *Trace* to follow it all the way back to the exact yarn lot.

PART G — MATERIAL REQUIREMENT: HOW THE QUANTITIES ARE CALCULATED

The recipe holds the rate *per unit of output*; the order multiplies it by the quantity and adds wastage. Three dosing bases (the Bangladesh conventions):

BASIS	FORMULA	EXAMPLE (ORDER SHOWN)
Per unit (yarn)	qty × rate × (1 + wastage)	400 × 1.0 × 1.05 = 420 kg yarn
% owf (dye)	qty × rate ÷ 100	460 × 3% = 13.8 kg dye
g/L (salt)	(qty × liquor) × rate ÷ 1000	460 × 8 × 40 ÷ 1000 = 147.2 kg salt
g/L (soda)	(qty × liquor) × rate ÷ 1000	460 × 8 × 15 ÷ 1000 = 55.2 kg soda

PRINTABLE DOCUMENTS (HAND TO THE FLOOR)

DOCUMENT	WHERE
Fabric spec sheet	Textile → Fabric Specifications → Spec sheet
Dyeing specification sheet	Textile → Dyeing Specifications → Recipe sheet
Lab-dip recipe sheet	Textile → Lab Dips → Recipe sheet
Knitting job card	Textile → Process Orders → Job card
Sub-contract work order	Textile → Knitting Sub-contracts → Work order
Production plan (T&A) sheet	Textile → Production Plans → Plan sheet

PART H — READY-MADE RECORDS TO TEST BY HAND

The demo also loads about **10 records in every module**, spread across their stages, so you can click each stage-specific button. Look for records marked **TC...** or noted **TESTCASE**.

MODULE	COUNT	STAGES YOU CAN TEST
Fabric Specifications	10	10 fabrics with recipe — spec sheet & auto-fill
Dyeing Specifications	10	10 dyed fabrics with dye/chemical recipe; 5 approved with recipe no + version

MODULE	COUNT	STAGES YOU CAN TEST
Process Routes	3	Knit→Dye→Finish, Knit→Finish, Dye→Compact (drive plan generation)
Fabric Rolls	3	Rolls generated on a completed run — pass/fail QC per roll
Rework runs	2	QC-reject runs each with a linked rework order
Lab Dips	10	Draft, Submitted, In-lab, Internal approved, Sent to customer, Customer approved, Rejected (+ Cancel action)
Process Orders	10	Draft, Planned (→ Issue), In-progress (→ Costs/Produce), QC (→ Record QC), Completed, Cancelled
Knitting Sub-contracts	10	Draft, Planned, In-progress (→ Record charge), Charged (→ Receive fabric), QC, Completed, Cancelled
Production Plans	10	Draft, Scheduled, In-progress, On-hold, Completed, Cancelled — each with buyer/style specs + stages

Re-load just these anytime with `php artisan db:seed --class="Database\Seeders\TextileTestCasesSeeder"` — it is idempotent (won't duplicate).

PART I — INTEGRATED ERP FEATURES (ADVANCED)

The textile module is an extension of the core ERP — it shares the same products, inventory, accounts, MRP, QC and approvals. These features connect it end to end:

FEATURE	WHAT IT DOES & WHERE
Textile product fields	Flag a product as textile (yarn/grey/dyed/finished) with GSM, width, colour, roll-tracked — on the Product form. Normal products are unaffected.
Wastage (expected vs actual)	Default wastage % per product & per process; each order shows expected output and, after production, actual wastage % & reason. See the Wastage report's Exp% / Act% columns.
Process routes	Textile → Process Routes — define a product's production sequence once; plans generated for that product follow the route.
Fabric rolls	Roll-tracked products split production into rolls (Textile → Fabric Rolls) with per-roll weight/GSM/QC; Roll Traceability report ties each roll to its batch and source materials.
Rework	On a completed run with QC rejects, <i>Create rework</i> raises a linked run (the original is never changed). Rework report lists them with added cost.
Versioned recipes	Dyeing/fabric specs carry recipe no + version + approval; recipes auto-scale to the order quantity. <i>Approve</i> to lock a version.
MRP → purchasing	Reports → MRP → <i>Create purchase requisition</i> turns shortages into a draft requisition in the normal purchasing flow.
Textile dashboard KPIs	Dashboard shows yarn/grey/dyed/finished stock, WIP, wastage %, pending QC, active runs and material shortages — live values.

TEST IT YOURSELF — 6-POINT CHECKLIST

#	CHECK	EXPECTED
1	New knitting order → pick Grey Fabric	Composition, GSM, dia 30, gauge 24, stitch length auto-fill.

#	CHECK	EXPECTED
2	Set qty 400 → <i>Calculate from recipe</i>	Cotton Yarn = 420 kg.
3	New dyeing order → pick Navy lab dip, qty 460 → Calculate	Dye 13.8 · Salt 147.2 · Soda 55.2 kg.
4	Knitting Sub-contracts → KNIT-0004	Charge Tk 50,000; grey received 1,960 kg.
5	Production Plans → PLAN-0002	Stages Knit → Dye → Finish.
6	Batches → open finished → <i>Trace</i>	Traces back to the yarn lot.

Every value above is live demo data in the DEMO company — open the screen and confirm it matches.

Thank you for your business.